

Real-Time Energy Demand, Generation and Electricity Price Analytics using Streaming Data and Apache Kafka

Streaming Data Analytics — Assignment 2



Industry:

Energy / Power Sector

Submitted By

Lakshay Gautam

Roll No.: 065069

PGDM — Big Data Analytics

Streaming Data Analytics

Assignment 2

**Industry | Data Sources | Data Pipeline| Streaming Data|
outputs**

AI Energy Control Tower

Real-Time Energy Intelligence & Streaming Analytics Platform Using Apache Kafka

Project Type

Streaming Data Analytics | Big Data Analytics | Real-Time Monitoring System

Technology Stack

Apache Kafka • Python • Docker • PostgreSQL • Open-Meteo API • Power BI • Stream Processing • Event-Driven Architecture

1. Executive Summary

The energy sector generates massive volumes of continuous data from power plants, transmission networks, electricity markets, weather systems, and renewable energy assets. Traditional batch-processing systems cannot provide the speed required for operational decision-making.

The AI Energy Control Tower was developed as a real-time streaming analytics platform capable of ingesting, processing, analyzing, and visualizing energy data as events occur.

The platform leverages Apache Kafka as the central streaming backbone to capture live data streams related to:

- Electricity Demand
- Power Generation
- Weather Conditions
- Electricity Prices
- Operational Alerts

The solution enables near real-time monitoring, anomaly detection, forecasting readiness, and executive decision support through a scalable event-driven architecture.

Key Business Objectives

- Monitor electricity demand in real time
- Track renewable and conventional generation
- Analyze weather impact on grid operations
- Detect anomalies instantly

- Enable future demand forecasting
 - Build a scalable Energy Intelligence Platform
-

2. Business Problem Statement

Energy operators face several challenges:

Challenge 1: Data Silos

Energy data originates from multiple sources:

- Weather Systems
- Generation Assets
- Grid Operators
- Energy Markets

These systems often operate independently.

Challenge 2: Delayed Decision Making

Traditional reporting systems process data in batches, causing delays in operational response.

Challenge 3: Renewable Energy Variability

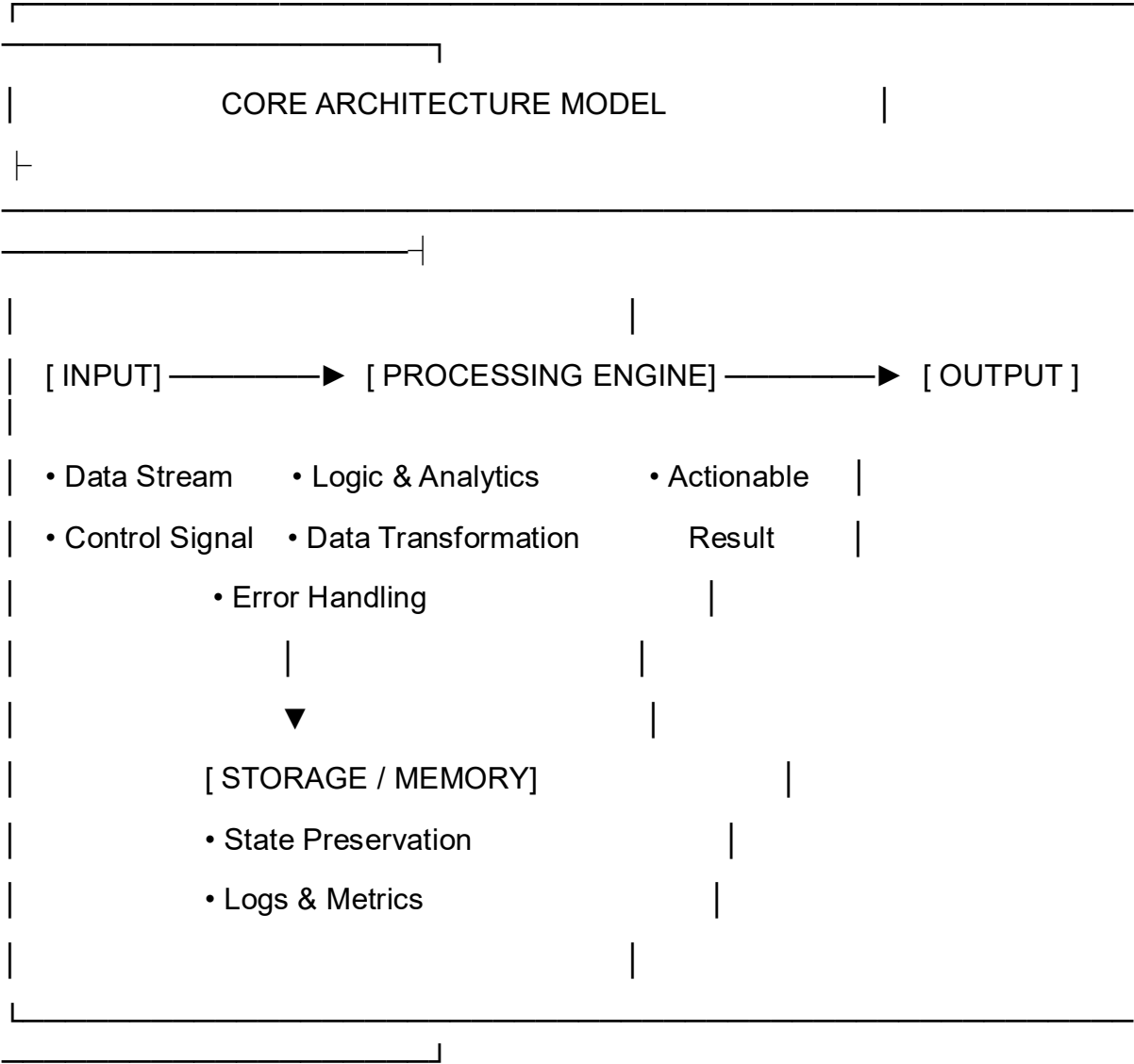
Solar and wind generation fluctuate rapidly due to changing weather conditions.

Challenge 4: Grid Reliability

Unexpected demand spikes or generation drops can affect grid stability.

Proposed Solution

Develop a centralized streaming platform capable of ingesting multiple data streams and delivering actionable insights in real time.



| Initialize | (Start Node)

+-----+-----+

|

v

+-----+

| Validate Inputs |

+-----+-----+

|

+-----+-----+

|

|

[Valid] [Invalid]

|

|

v

v

+-----+ +-----+

| Process Data || Raise Error |

+-----+-----+ +-----+-----+

|

|

+-----+-----+

|

v

+-----+

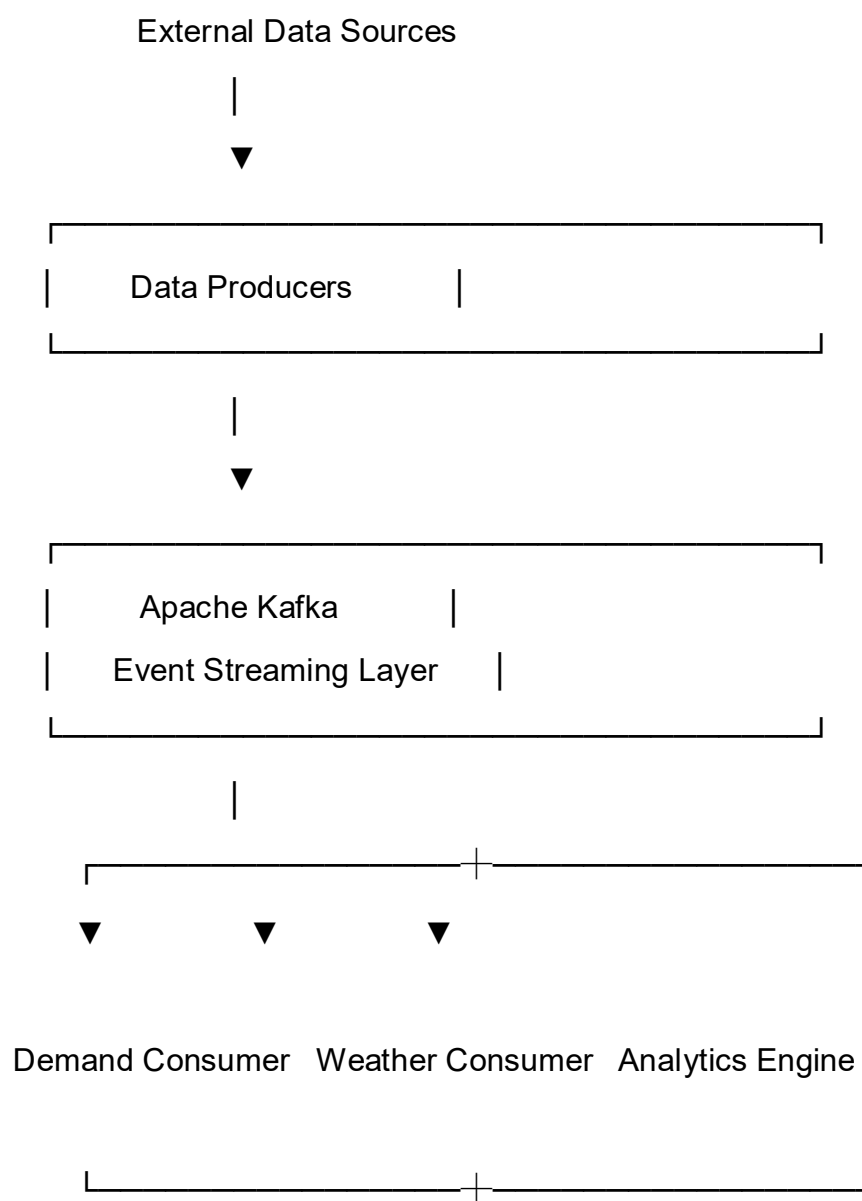
| Generate |

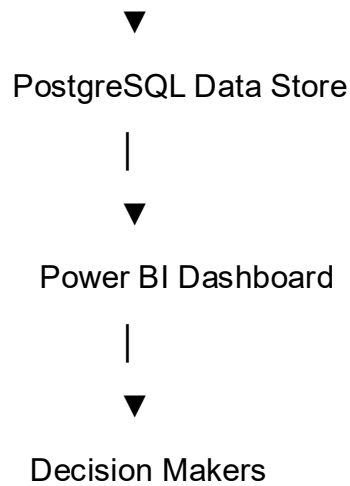
| Final Output | (End Node)

+-----+

3. Solution Architecture

High-Level System Architecture





Architectural Style

The platform follows an Event-Driven Architecture (EDA).

Benefits:

- Real-time processing
- Loose coupling
- High scalability
- Fault tolerance
- Independent producer-consumer model

4. Data Streaming Architecture

The system uses Apache Kafka as a distributed event streaming platform.

Kafka Cluster

Kafka acts as the central message broker.

Verified Kafka Topics:

Topic	Purpose
weather-data	Weather Streaming
grid-demand	Electricity Demand
generation-data	Power Generation
price-data	Market Prices
anomaly-alerts	Operational Alerts

Streaming Workflow

Data Source

|



Producer

|



Kafka Topic

|



Consumer

|



Analytics Layer

|



Dashboard

Why Kafka?

- High Throughput
 - Distributed Processing
 - Real-Time Streaming
 - Horizontal Scalability
 - Event Persistence
-

```
-----
🔥 STARTING OPEN-METEO WEATHER PRODUCER
-----
✅ Open-Meteo producer started (PID: 106689)

-----
🇮🇳 STARTING SYNTHETIC INDIA ENERGY PRODUCER
-----
✅ Synthetic producer started (PID: 106716)

Waiting for producers to initialize...

-----
📺 STARTING REAL-TIME DATA DISPLAY
-----
✅ Display consumer started (PID: 106758)

=====
✅ ALL COMPONENTS STARTED SUCCESSFULLY!
=====

📡 Streaming data from:
  ✓ Open-Meteo (Weather Data)
  ✓ Synthetic India (Energy Demand & Generation)

🖥️ Display is live on the console above

🛑 Press Ctrl+C to stop all services
```

```
=====
🔥 AI ENERGY CONTROL TOWER - STREAMING DATA PIPELINE 🔥
=====

Streaming all data sources through Kafka in real-time

=====
📺 STREAMING CONFIGURATION
=====
Bootstrap Servers:  localhost:9092
Producers:          Open-Meteo Weather + Synthetic India Energy
Data Topics:        weather-data, grid-demand, generation-data, price-data, anomaly-alerts
Display Mode:        Real-time console with colored output
Duration:            Continuous (until Ctrl+C)
=====

🔥 Launching all producers...

-----
🔥 STARTING OPEN-METEO WEATHER PRODUCER
-----
✅ Open-Meteo producer started (PID: 106689)

-----
🇮🇳 STARTING SYNTHETIC INDIA ENERGY PRODUCER
```

```
> __pycache__
> analytics
> backend
> __pycache__
> api
> config
> database
> models
> schemas
> services
> utils
> _init_.py
> main.py
> config
> dashboard
> data
> database
> docker
> docs
> energy_kafka
> logs
> notebooks
> Outline
> Timeline

\venv\scripts\python\python.exe -c "python energy_control_tower\docker_exe.py --it energy_kafka /opt/kafka/bin/kafka-console-consumer.sh \
--bootstrap-server localhost:19094 \
--topic price-data \
--from-beginning"

{"timestamp": "2026-09-01T14:40:27.420525+00:00", "region": "Northeast", "price_per_unit": 4.6695357531258015, "currency": "INR", "market_type": "DAM", "volume_mwh": 4417.08, "volatility_percent": 5.95, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:32.838403+00:00", "region": "West", "price_per_unit": 3.928774199733577, "currency": "INR", "market_type": "DAM", "volume_mwh": 2570.13, "volatility_percent": 3.25, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:32.838405+00:00", "region": "Northeast", "price_per_unit": 4.096719203911026, "currency": "INR", "market_type": "DAM", "volume_mwh": 1561.82, "volatility_percent": 3.87, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:37.910275+00:00", "region": "West", "price_per_unit": 4.15020367561631, "currency": "INR", "market_type": "DAM", "volume_mwh": 4523.45, "volatility_percent": 4.34, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:37.910279+00:00", "region": "Northeast", "price_per_unit": 4.249131565979097, "currency": "INR", "market_type": "DAM", "volume_mwh": 3560.16, "volatility_percent": 7.18, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:43.051572+00:00", "region": "West", "price_per_unit": 4.725024771132164, "currency": "INR", "market_type": "DAM", "volume_mwh": 4670.58, "volatility_percent": 2.21, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:43.051577+00:00", "region": "Northeast", "price_per_unit": 4.651298102396714, "currency": "INR", "market_type": "DAM", "volume_mwh": 1558.28, "volatility_percent": 3.14, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282703+00:00", "region": "West", "price_per_unit": 4.641540161515536, "currency": "INR", "market_type": "DAM", "volume_mwh": 4533.15, "volatility_percent": 3.28, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282708+00:00", "region": "Northeast", "price_per_unit": 4.474927656337997, "currency": "INR", "market_type": "DAM", "volume_mwh": 2821.6, "volatility_percent": 3.18, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:12:18.966917+00:00", "region": "South", "price_per_unit": 4.78, "currency": "INR", "market_type": "DAM", "volume_mwh": 1217.0, "volatility_percent": 3.35, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:12:18.967169+00:00", "region": "West", "price_per_unit": 4.96, "currency": "INR", "market_type": "DAM", "volume_mwh": 1324.0, "volatility_percent": 4.72, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:12:18.967284+00:00", "region": "West", "price_per_unit": 4.43, "currency": "INR", "market_type": "DAM", "volume_mwh": 3852.0, "volatility_percent": 7.92, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:12:18.967421+00:00", "region": "West", "price_per_unit": 4.25, "currency": "INR", "market_type": "DAM", "volume_mwh": 2962.0, "volatility_percent": 5.52, "data_source": "synthetic"}
```

```

(venv) lakshay@Lakshaygtm-27:~/AI-Energy-Control-Tower$ docker exec -it energy-kafka /opt/kafka/bin/kafka-console-consumer.sh \
--bootstrap-server localhost:19094 \
--topic generation-data \
--from-beginning
{"timestamp": "2026-09-01T14:40:48.282494+00:00", "region": "South", "fuel_type": "Solar", "generation_mw": 2231.23, "capacity_mw": 2789.04, "utilization_percent": 80, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282499+00:00", "region": "South", "fuel_type": "Wind", "generation_mw": 1216.57, "capacity_mw": 1520.71, "utilization_percent": 80, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282504+00:00", "region": "South", "fuel_type": "Nuclear", "generation_mw": 644.1, "capacity_mw": 805.12, "utilization_percent": 80, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282508+00:00", "region": "South", "fuel_type": "Biomass", "generation_mw": 2147.0, "capacity_mw": 2683.75, "utilization_percent": 80, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282513+00:00", "region": "East", "fuel_type": "Coal", "generation_mw": 9661.5, "capacity_mw": 12076.88, "utilization_percent": 80, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282518+00:00", "region": "East", "fuel_type": "Natural Gas", "generation_mw": 1717.6, "capacity_mw": 2147.0, "utilization_percent": 80, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282523+00:00", "region": "East", "fuel_type": "Hydro", "generation_mw": 3220.5, "capacity_mw": 4025.62, "utilization_percent": 80, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282530+00:00", "region": "East", "fuel_type": "Solar", "generation_mw": 2231.23, "capacity_mw": 2789.04, "utilization_percent": 80, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282535+00:00", "region": "East", "fuel_type": "Wind", "generation_mw": 766.84, "capacity_mw": 958.55, "utilization_percent": 80, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282540+00:00", "region": "East", "fuel_type": "Nuclear", "generation_mw": 644.1, "capacity_mw": 805.12, "utilization_percent": 80, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282545+00:00", "region": "East", "fuel_type": "Biomass", "generation_mw": 2147.0, "capacity_mw": 2683.75, "utilization_percent": 80, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:11:39.890293+00:00", "region": "North", "fuel_type": "Biomass", "generation_mw": 2147.0, "capacity_mw": 2683.8, "utilization_percent": 80, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:12:18.751609+00:00", "region": "South", "fuel_type": "Hydro", "generation_mw": 3220.5, "capacity_mw": 4025.6, "utilization_percent": 80, "data_source": "synthetic"}

```

```

(venv) lakshay@Lakshaygtm-27:~/AI-Energy-Control-Tower$ docker exec -it energy-kafka /opt/kafka/bin/kafka-console-consumer.sh \
--bootstrap-server localhost:19094 \
--topic grid-demand \
--from-beginning
ay_of_week": 1, "is_holiday": false, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:27.420365+00:00", "region": "East", "demand_mw": 18674.96, "peak_demand_mw": 30000.0, "hour_of_day": 14, "day_of_week": 1, "is_holiday": false, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:32.838249+00:00", "region": "North", "demand_mw": 33535.6, "peak_demand_mw": 52500.0, "hour_of_day": 14, "day_of_week": 1, "is_holiday": false, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:32.838269+00:00", "region": "South", "demand_mw": 23300.09, "peak_demand_mw": 37500.0, "hour_of_day": 14, "day_of_week": 1, "is_holiday": false, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:32.838272+00:00", "region": "East", "demand_mw": 18579.41, "peak_demand_mw": 30000.0, "hour_of_day": 14, "day_of_week": 1, "is_holiday": false, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:37.909932+00:00", "region": "North", "demand_mw": 31840.55, "peak_demand_mw": 52500.0, "hour_of_day": 14, "day_of_week": 1, "is_holiday": false, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:37.909969+00:00", "region": "South", "demand_mw": 24199.63, "peak_demand_mw": 37500.0, "hour_of_day": 14, "day_of_week": 1, "is_holiday": false, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:37.909976+00:00", "region": "East", "demand_mw": 19904.19, "peak_demand_mw": 30000.0, "hour_of_day": 14, "day_of_week": 1, "is_holiday": false, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:43.051145+00:00", "region": "North", "demand_mw": 32136.4, "peak_demand_mw": 52500.0, "hour_of_day": 14, "day_of_week": 1, "is_holiday": false, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:43.051213+00:00", "region": "South", "demand_mw": 23365.93, "peak_demand_mw": 37500.0, "hour_of_day": 14, "day_of_week": 1, "is_holiday": false, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:43.051221+00:00", "region": "East", "demand_mw": 18107.61, "peak_demand_mw": 30000.0, "hour_of_day": 14, "day_of_week": 1, "is_holiday": false, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282330+00:00", "region": "North", "demand_mw": 34210.15, "peak_demand_mw": 52500.0, "hour_of_day": 14, "day_of_week": 1, "is_holiday": false, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282369+00:00", "region": "South", "demand_mw": 23533.72, "peak_demand_mw": 37500.0, "hour_of_day": 14, "day_of_week": 1, "is_holiday": false, "data_source": "synthetic"}
{"timestamp": "2026-09-01T14:40:48.282378+00:00", "region": "East", "demand_mw": 18756.78, "peak_demand_mw": 30000.0, "hour_of_day": 14, "day_of_week": 1, "is_holiday": false, "data_source": "synthetic"}

```

```

(venv) lakshay@Lakshaygtm-27:~/AI-Energy-Control-Tower$ docker exec -it energy-kafka /opt/kafka/bin/kafka-console-consumer.sh \
--bootstrap-server localhost:19094 \
--topic weather-data \
--from-beginning
{"timestamp": "2026-09-01T14:12:18.857531+00:00", "location": "Mumbai", "region": "West", "temperature_celsius": 30.1, "humidity_percent": 74.7, "wind_speed_kmh": 44.4, "weather_code": 95, "data_source": "open-meteo"}
{"timestamp": "2026-09-01T14:40:29.044539+00:00", "location": "Bangalore", "latitude": 12.9716, "longitude": 77.5946, "temperature_celsius": 22.7, "apparent_temperature_celsius": 25.5, "humidity_percent": 85, "precipitation_mm": 0.0, "wind_speed_kmh": 7.5, "wind_direction_degrees": 253, "cloud_cover_percent": 34, "weather_code": 1, "data_source": "open-meteo", "region": "South"}
{"timestamp": "2026-09-01T14:40:30.420266+00:00", "location": "Ahmedabad", "latitude": 23.0225, "longitude": 72.5714, "temperature_celsius": 32.7, "apparent_temperature_celsius": 36.2, "humidity_percent": 50, "precipitation_mm": 0.0, "wind_speed_kmh": 4.9, "wind_direction_degrees": 213, "cloud_cover_percent": 58, "weather_code": 2, "data_source": "open-meteo", "region": "West"}
{"timestamp": "2026-09-01T14:40:30.616825+00:00", "location": "Jaipur", "latitude": 26.9124, "longitude": 75.7873, "temperature_celsius": 28.7, "apparent_temperature_celsius": 32.8, "humidity_percent": 67, "precipitation_mm": 0.0, "wind_speed_kmh": 5.2, "wind_direction_degrees": 243, "cloud_cover_percent": 100, "weather_code": 3, "data_source": "open-meteo", "region": "North"}
{"timestamp": "2026-09-01T14:41:33.129822+00:00", "location": "Bangalore", "latitude": 12.9716, "longitude": 77.5946, "temperature_celsius": 22.7, "apparent_temperature_celsius": 25.5, "humidity_percent": 85, "precipitation_mm": 0.0, "wind_speed_kmh": 7.5, "wind_direction_degrees": 253, "cloud_cover_percent": 34, "data_source": "open-meteo", "region": "South"}
{"timestamp": "2026-09-01T14:41:34.177683+00:00", "location": "Ahmedabad", "latitude": 23.0225, "longitude": 72.5714, "temperature_celsius": 32.7, "apparent_temperature_celsius": 36.2, "humidity_percent": 50, "precipitation_mm": 0.0, "wind_speed_kmh": 4.9, "wind_direction_degrees": 213, "cloud_cover_percent": 58, "weather_code": 2, "data_source": "open-meteo", "region": "West"}
{"timestamp": "2026-09-01T14:41:34.389898+00:00", "location": "Jaipur", "latitude": 26.9124, "longitude": 75.7873, "temperature_celsius": 28.7, "apparent_temperature_celsius": 32.8, "humidity_percent": 67, "precipitation_mm": 0.0, "wind_speed_kmh": 5.2, "wind_direction_degrees": 243, "cloud_cover_percent": 100, "weather_code": 3, "data_source": "open-meteo", "region": "North"}

```

```
> __pycache__
> api
> config
> database
> models
> schemas
> services
> utils
+ _init_.py
+ main.py
> config
> dashboard
> data
> database
> docker
> docs
+ energy_kafka
> logs
> notebooks
> Outline
> Timeline

(venv) lakshay@Lakshaytm-27:~/AI-Energy-Control-Tower$ docker exec -it energy-kafka /opt/kafka/bin/kafka-console-consume
r.sh \
--bootstrap-server localhost:19094 \
--topic weather-data \
--from-beginning
{"timestamp": "2026-09-01T14:40:30.420266+00:00", "location": "Ahmedabad", "latitude": 23.0225, "longitude": 72.5714, "temperature_celsius": 32.7, "apparent_temperature_celsius": 36.2, "humidity_percent": 50, "precipitation_mm": 0.0, "wind_speed_kmh": 4.9, "wind_direction_degrees": 213, "cloud_cover_percent": 58, "weather_code": 2, "data_source": "open-meteo", "region": "West"}
{"timestamp": "2026-09-01T14:40:30.616825+00:00", "location": "Jaipur", "latitude": 26.9124, "longitude": 75.7873, "temperature_celsius": 28.7, "apparent_temperature_celsius": 32.8, "humidity_percent": 67, "precipitation_mm": 0.0, "wind_speed_kmh": 5.2, "wind_direction_degrees": 243, "cloud_cover_percent": 100, "weather_code": 3, "data_source": "open-meteo", "region": "North"}
{"timestamp": "2026-09-01T14:41:33.129822+00:00", "location": "Bangalore", "latitude": 12.9716, "longitude": 77.5946, "temperature_celsius": 22.7, "apparent_temperature_celsius": 25.5, "humidity_percent": 85, "precipitation_mm": 0.0, "wind_speed_kmh": 7.5, "wind_direction_degrees": 253, "cloud_cover_percent": 34, "weather_code": 1, "data_source": "open-meteo", "region": "South"}
{"timestamp": "2026-09-01T14:41:34.177683+00:00", "location": "Ahmedabad", "latitude": 23.0225, "longitude": 72.5714, "temperature_celsius": 32.7, "apparent_temperature_celsius": 36.2, "humidity_percent": 50, "precipitation_mm": 0.0, "wind_speed_kmh": 4.9, "wind_direction_degrees": 213, "cloud_cover_percent": 58, "weather_code": 2, "data_source": "open-meteo", "region": "West"}
{"timestamp": "2026-09-01T14:41:34.389898+00:00", "location": "Jaipur", "latitude": 26.9124, "longitude": 75.7873, "temperature_celsius": 28.7, "apparent_temperature_celsius": 32.8, "humidity_percent": 67, "precipitation_mm": 0.0, "wind_speed_kmh": 5.2, "wind_direction_degrees": 243, "cloud_cover_percent": 100, "weather_code": 3, "data_source": "open-meteo", "region": "North"}

energy_kafka/producers/synthetic_india_producer.py: self.renewable_topic = renewable_topic
energy_kafka/producers/synthetic_india_producer.py: self.price_topic = price_topic
energy_kafka/producers/synthetic_india_producer.py: self.alerts_topic = alerts_topic
energy_kafka/producers/synthetic_india_producer.py: represented on the anomaly-alerts topic even in short runs.
energy_kafka/producers/synthetic_india_producer.py: # Always emit one randomized alert per cycle so the topic is populated
energy_kafka/producers/synthetic_india_producer.py: """Generate and produce all types of data to their aligned topics"""
energy_kafka/producers/synthetic_india_producer.py: # Send to respective topics (send_message / send_message_to_topic)
energy_kafka/producers/synthetic_india_producer.py: self.send_message_to_topic(self.generation_topic, msg, key=msg["region"])
energy_kafka/producers/synthetic_india_producer.py: self.send_message_to_topic(self.price_topic, msg, key=msg["region"])
energy_kafka/producers/synthetic_india_producer.py: self.send_message_to_topic(self.alerts_topic, msg, key=msg["region"])
energy_kafka/producers/openmeteo_producer.py: topic: str = "weather-data",
energy_kafka/producers/openmeteo_producer.py: topic: Kafka topic for weather data
energy_kafka/producers/openmeteo_producer.py: topic=topic,
energy_kafka/producers/energy_producer.py: print("Streaming data to 'energy_data' topic...")
energy_kafka/producers/openmeteo_producer.py: grep: energy_kafka/producers/___pycache___/synthetic_india_producer.cpython-314.pyc: binary file matches
energy_kafka/producers/openmeteo_producer.py: grep: energy_kafka/producers/___pycache___/openmeteo_producer.cpython-314.pyc: binary file matches
energy_kafka/producers/openmeteo_producer.py: grep: energy_kafka/producers/___pycache___/base_producer.cpython-314.pyc: binary file matches
energy_kafka/producers/openmeteo_producer.py: grep: energy_kafka/producers/___pycache___/energy_producer.cpython-314.pyc: binary file matches
energy_kafka/producers/base_producer.py: topic: str,
energy_kafka/producers/base_producer.py: topic: Kafka topic to publish to
energy_kafka/producers/base_producer.py: self.topic = topic
energy_kafka/producers/base_producer.py: Send message to Kafka topic
energy_kafka/producers/base_producer.py: self.topic,
energy_kafka/producers/base_producer.py: f"Message sent to {record_metadata.topic}:"
energy_kafka/producers/base_producer.py: def send_message_to_topic(
energy_kafka/producers/base_producer.py: self, topic: str, message: Dict[str, Any], key: Optional[str] = None
energy_kafka/producers/base_producer.py: Send a message to an arbitrary topic (not only self.topic).
energy_kafka/producers/base_producer.py: topic: Target Kafka topic
energy_kafka/producers/base_producer.py: topic,
energy_kafka/producers/base_producer.py: f"Message sent to {record_metadata.topic}:"
energy_kafka/producers/base_producer.py: logger.error(f"Kafka error sending message to {topic}: {e}")
```

5. Data Sources and Ingestion Pipeline

Weather Data Stream

Source

Open-Meteo API

Topic

weather-data

Key Attributes

- Temperature
- Humidity
- Wind Speed
- Cloud Cover

Business Impact

Weather significantly influences electricity demand and renewable energy generation.

Insert Screenshot: Weather Producer Output

Grid Demand Stream

Topic

grid-demand

Key Attributes

- Region
- Demand (MW)
- Peak Load
- Timestamp

Business Impact

Provides visibility into power consumption patterns.

Insert Screenshot: Demand Data Stream

Generation Stream

Topic

generation-data

Key Attributes

- Thermal Generation
- Solar Generation
- Wind Generation
- Hydro Generation

Business Impact

Tracks energy production mix and renewable penetration.

Insert Screenshot: Generation Stream Messages

Electricity Price Stream

Topic

price-data

Key Attributes

- Market Price
- Region
- Timestamp

Business Impact

Supports market intelligence and price forecasting.

Insert Screenshot: Price Data Stream

Anomaly Alert Stream

Topic

anomaly-alerts

Purpose

Generate alerts when abnormal conditions occur.

Examples:

- Demand Spike
- Generation Drop
- Weather Event
- Price Surge

Insert Screenshot: Alert Messages

6. Technical Implementation

Producer Layer

Location:

energy_kafka/producers

Components:

OpenMeteo Producer

Collects weather information from APIs.

Synthetic India Producer

Generates simulated Indian power grid demand data.

Energy Producer

Publishes generation and market events.

Consumer Layer

Location:

energy_kafka/consumers

Responsibilities:

- Subscribe to Kafka topics
 - Process incoming events
 - Validate messages
 - Store processed records
-

Storage Layer

PostgreSQL

Purpose:

- Historical Data Storage
- Analytics Data Repository
- Dashboard Data Source

Benefits:

- Structured Storage
 - Query Optimization
 - Historical Analysis
-

7. Analytics Framework

The analytics layer converts raw events into actionable insights.

Demand Analytics

Tracks:

- Current Demand
- Peak Demand

- Demand Trends
-

Generation Analytics

Tracks:

- Generation Mix
 - Renewable Contribution
 - Capacity Utilization
-

Price Analytics

Tracks:

- Price Trends
 - Volatility
 - Market Behaviour
-

Weather Analytics

Tracks:

- Temperature Impact
 - Weather Correlation
 - Renewable Forecast Inputs
-

Anomaly Detection

Detects:

- Demand Outliers
 - Generation Failures
 - Price Spikes
 - Weather Extremes
-

8. Infrastructure Deployment

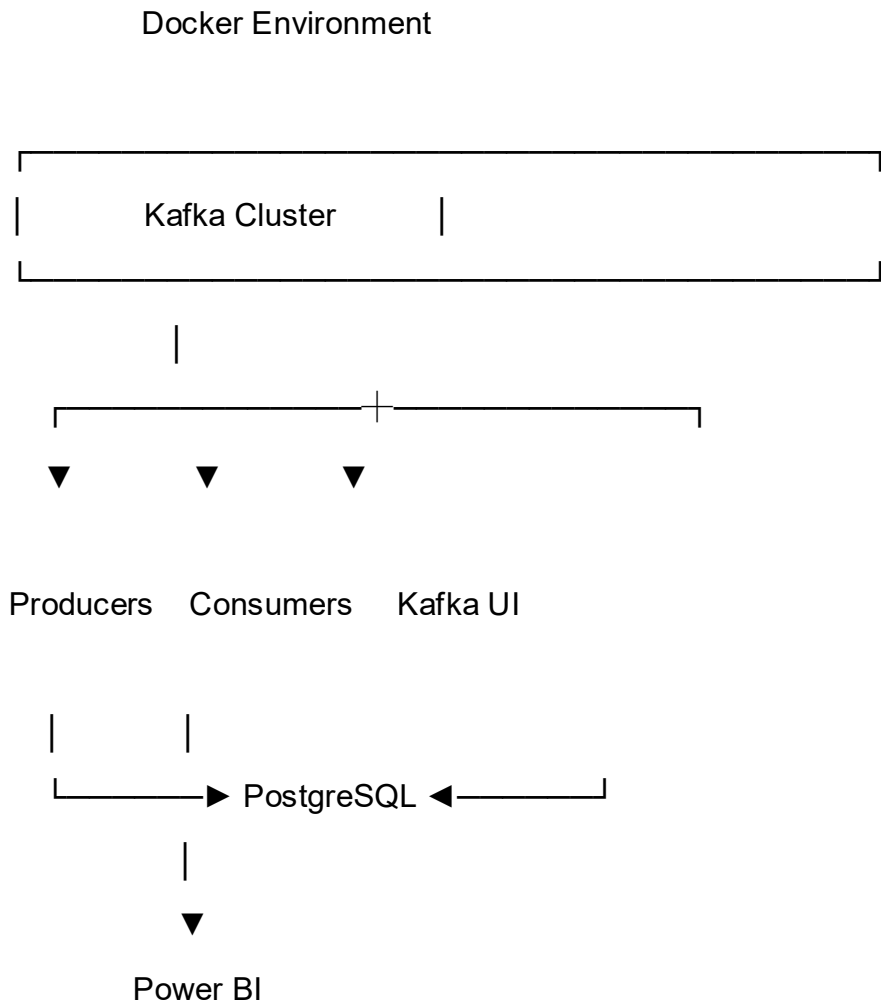
Containerized Architecture

Docker was used to deploy and manage services.

Running Containers

Container	Purpose
energy-kafka	Event Streaming
energy-kafka-ui	Kafka Monitoring
energy-postgres	Data Storage
MongoDB	NoSQL Storage
n8n	Workflow Automation
Ollama	AI Integration

Infrastructure Diagram



Insert Screenshot: Docker Containers

Insert Screenshot: Kafka UI

9. Results and Outcomes

The platform successfully demonstrated:

Real-Time Streaming

Continuous event ingestion through Kafka topics.

Multi-Source Integration

Weather, demand, generation, and pricing data unified into a single platform.

Event Processing

Streaming data processed with low latency.

Monitoring Capability

Centralized operational visibility through dashboards.

Scalability

Architecture supports future expansion to additional energy sources and analytics workloads.

10. Future Enhancements

Phase 2

- Live GRID-INDIA Integration
- CEA Real-Time Feeds
- Renewable Forecasting Models
- LSTM Demand Forecasting
- Predictive Maintenance

Phase 3

- AI Copilot using Ollama
- GenAI-Based Energy Insights
- Automated Alerting using n8n
- Smart Grid Digital Twin
- Carbon Emission Analytics